4.4 Waves

This section provides knowledge and understanding of wave properties, electromagnetic waves, superposition and stationary waves. The wavelength of visible light is too small to be measured directly using a ruler. However, superposition experiments can be done in the laboratory to determine wavelength of visible light using a laser and a double slit.

There are opportunities to discuss how the double-slit experiment demonstrated the wave-like behaviour of light (HSW7).

The breadth of the topic covering sound waves and the electromagnetic spectrum provides scope for learners to appreciate the wide ranging applications of waves and their properties. (HSW1, 2, 5, 8, 9, 12)

4.4.1 Wave motion

		Learning outcomes	Additional guidance
		Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)		progressive waves; longitudinal and transverse waves	HSW8
(b)	**(i)**	displacement, amplitude, wavelength, period, phase difference, frequency and speed of a wave	HSW8
	(ii)	techniques and procedures used to use an oscilloscope to determine frequency	**PAG5**
(c)		the equation $f = \frac{1}{T}$	
(d)		the wave equation $v = f\lambda$	

(e)	graphical representations of transverse and longitudinal waves		HSW5
(f)	**(i)**	reflection, refraction, polarisation and diffraction of all waves	Learners will be expected to know that diffraction effects become significant when the wavelength is comparable to the gap width.
	(ii)	techniques and procedures used to demonstrate wave effects using a ripple tank	HSW1, 4
	(iii)	techniques and procedures used to observe polarising effects using microwaves and light	**PAG5**
(g)	intensity of a progressive wave; $I = \frac{P}{A}$; intensity $\propto$ (amplitude)^2.		